Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int floorValue(int a[], int left, int right, int x) {
    if (left > right)
        return -1;

    int mid = (left + right) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] > x)
        return floorValue(a, left, mid - 1, x);

    int result = floorValue(a, mid + 1, right, x);

    if (result == -1)
        return a[mid];

    return result;
}

int main() {
    int n, x;
    scanf("%d", &n);

    int a[n];
    int i = 0;

    while (i < n) {
        scanf("%d", &a[i]);
        i = i + 1;
    }

    scanf("%d", &x);

    printf("%d", floorValue(a, 0, n - 1, x));

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.